

Matthew J. Cockayne

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ABOUT ME

PhD researcher in safe and responsible AI, working on AI systems that are interpretable, robust, and equitable. My work includes detecting and mitigating bias, stress-testing models under real-world distribution shift, and building interpretable-by-design architectures. Applied primarily to medical imaging, my work has produced robust, fairness-improved, and clinically auditable models, published across journals and conferences, including an early-accepted (top 9%) MICCAI 2026 paper. I am currently extending this research work to validate vision-language models in cardiology. I bring research experience and hands-on ability to translate this into reproducible, well-documented tools and materials for interdisciplinary researchers.

SKILLS

Programming: Python, R, SQL, Git, LaTeX, Weights & Biases, TensorBoard

ML & Computer Vision: PyTorch, TensorFlow, scikit-learn, Hugging Face, OpenCV, TIMM, torchvision

Frontier AI: Vision-Language Models (VLMs), SAM/MedSAM, Zero-Shot Evaluation, Failure Mode Analysis

AI Safety & XAI: Interpretability methods, Fairness/Robustness Benchmarking, Uncertainty Quantification, SHAP, LIME, CAM-methods, LRP, Concept Bottleneck Models

Data & HPC: Pandas, NumPy, SciPy, SLURM Job Scheduling, Distributed LLM/VLM Training & Inference

Research: Literature Reviews, Scientific Writing, Experiment Design, Publication, Grant Writing, peer review, Bias/Fairness Auditing, Model Interpretability Assessment

Certifications: BlueDot Technical AI Safety Course

EXPERIENCE

Research Assistant, Vision-Language Models in Cardiology Reporting 2026-Present
Keele University, School of Computer Science and Mathematics

- Investigating whether vision-language models for cardiology reporting learn clinically valid internal representations or instead exploit spurious correlations (scanner artefacts, demographics)
- Validation of model representations in collaboration with a practising cardiologist

Lead Session Teacher, Level 7 Apprenticeship (NHS Partnership) 2026-Present
Keele University, School of Chemical and Physical Sciences

- Leading delivery of the image processing module on Keele's Level 7 technical leadership apprenticeship programme, delivered in partnership with the NHS for practising healthcare and technology professionals.

PhD Researcher, Safe & Responsible AI for Medical Imaging 2023-Present
Keele University, School of Computer Science and Mathematics

- Published a robustness evaluation framework benchmarking transformer architectures under real-world corruptions. Applied to multi-modal skin cancer detection, achieving state-of-the-art performance while quantifying deployment risk via uncertainty estimation.

Code: github.com/xraikeye/DermFormer

- Developed SkinCBM, an interpretable-by-design concept bottleneck model aligned to the 7-point checklist for melanoma, enabling clinicians to inspect and correct intermediate reasoning steps.

Code: github.com/Matt-Cockayne/SkinCBM

- Conducted information analysis of learnt skin lesion concepts, revealing reliance on spurious correlations, and designed sparse polynomial classifiers that preserve full interpretability of decision boundaries (under-review, Image and Vision Computing).

- Designed FairCBM, a fairness-aware curriculum learning scheme for concept bottleneck models in dermatology, reducing demographic performance disparities by 44% and improving lowest-group outcomes by 63%. Early accepted (top 9%) at MICCAI 2026.

Code: github.com/Matt-Cockayne/FairCBM

- Evaluated frontier model reliability for skin lesion segmentation under zero-shot transfer, characterising failure modes and developing adaptive selection methods that reduce catastrophic failures.

Code: github.com/xraikeye/Classification-to-Segmentation

- Authored and delivered tutorials on distributed VLM/LLM training and inference on Keele's GreenHPC cluster, supporting large-scale experimentation across research groups.

Code: github.com/Matt-Cockayne/keele-greenhpc-llm

Laboratory Demonstrator 2022-Present
Keele University, School of Computer Science and Mathematics

- Supported undergraduate and master's students across computer science modules for 4 years.
- Delivered guest lectures on deep learning, computer vision, and responsible AI.
- Mentored students on machine learning and AI ethics research projects.

Turing Network Development Research Associate 2021-2022

- Developed educational materials, organised networking events, and set-up hackathons to foster interdisciplinary research across the university as part of a grant from the Alan Turing Institute.

EDUCATION

PhD in Computer Science

2023-2026

Keele University, Keele, UK

Thesis: Responsible AI for Healthcare: Robustness, Interpretability, and Fairness in Skin Lesion Recognition

Viva: Passed with Minor Corrections

Funding: Competition-funded Department Studentship

Published two first-author papers (Pattern Analysis and Applications journal, AIIH 2025 conference, oral presentation) and one first-author paper early accepted (top 9%) at MICCAI 2026. One additional first-author paper under-review at Image and Vision Computing. Co-author on an accepted Discover Computing journal paper, an accepted AIIH 2026 paper and a Machine Vision and Applications paper under review. Developed open-source repositories: MedXAI, SkinCBM, and FairCBM for reproducible research. PGR Representative for 3 years.

MSc in Artificial Intelligence and Data Science, Distinction

2020-2021

Keele University, Keele, UK

Touchscreen Talent Data Analyst Internship: Developed explainable AI methods for CV-based sentiment analysis in recruitment.

BSc in Physics, First Class Honours

2017-2020

Keele University, Keele, UK

Undergraduate student representative for 3 years.

PUBLICATIONS

Journal Articles

- **Cockayne, M. J.**, Ortolani, M., Al-Bander, B. (2025). "DermFormer: Nested Multi-modal Vision Transformers for Robust Skin Cancer Detection." *Pattern Analysis and Applications*, Springer Nature, Volume 28, Article 194. DOI: [10.1007/s10044-025-01572-0](https://doi.org/10.1007/s10044-025-01572-0)

- **Cockayne, M. J.**, Ortolani, M., Al-Bander, B. "Sparse Concept Absence Network for Concept Bottleneck Models." *Image and Vision Computing*, Elsevier. Status: Under Review.

- Al-Bander, B., Ortolani, M., Dafaalla, M., **Cockayne, M. J.**, Pergola, G., Mamas, M. A. "Investigating and Mitigating Bias in Cardiovascular Disease Mortality Predictors for Scaling-up AI Fairness." *Discover Computing*, Springer Nature, 29, 365 (2026). DOI: doi.org/10.1007/s10791-026-10290-y

- Guo, S., **Cockayne, M. J.**, Ortolani, M., Al-Hamd, R., Al-Bander, B. "Zero-Shot Crack Segmentation for Structural Health Monitoring Using Class Activation Map Guided Segment Anything Model." *Machine Vision and Applications*. Status: Under Review.

Conference Papers

- **Cockayne, M. J.**, Ortolani, M., Al-Bander, B. (2025). "Classification-to-Segmentation: Class Activation Mapping for Zero-Shot Skin Lesion Segmentation." International Conference on AI in Healthcare (*AIiH 2025*), University of Cambridge. [Oral Presentation] DOI: [10.1007/978-3-032-00656-1_24](https://doi.org/10.1007/978-3-032-00656-1_24)

- **Cockayne, M. J.**, Ortolani, M., Al-Bander, B. "Fair Curriculum Learning for Concept Bottleneck Models in Dermatology." 29th International Conference on Medical Image Computing and Computer Assisted Intervention (*MICCAI 2026*). Status: Early Accepted Paper (Top 9%) & Accepted for Oral Presentation. Code: github.com/Matt-Cockayne/FairCBM

- Raza, A., **Cockayne, M. J.**, Ortolani, M., Hill, J., Al-Bander, B. "ClinAuditAI: Agentic AI for Clinical Guideline Adherence Auditing." International Conference on AI in Healthcare (*AIiH 2026: Explainability to Accountability*). DOI: doi.org/10.1007/978-3-032-35390-0_9

PROFESSIONAL SERVICE

Programme Committee: International Conference on AI in Healthcare (AIiH) 2026

Special Session Organiser: Co-organising a special session on explainability and accountability at AIiH 2026

Peer Review:

- Engineering Applications of Artificial Intelligence, Elsevier

- IET Computer Vision, Wiley

- 1st International Workshop on AI Safety and Security (AI-SS)

- 3rd International Conference on AI in Healthcare (AIiH)

Leadership: Postgraduate Research Representative, Keele University (2022-2025). Student Representative, Physics Programme (2018-2021)

Mentorship: Supervised student research projects on ML and AI, responsible AI, and fairness evaluation

Outreach:

- Keele & Sanda University AI Summer School (2026), Developed and co-facilitated a week program of ML content for visiting International students.

- Data Science Content Creator Internship (2021-2022), developed AI/data science educational resources.

Communication: Departmental research seminars and guest lectures on safe and responsible AI, robustness, fairness-aware learning, and interpretability (2022-present)